Supplementary Information (Online Resource 2)

Article title: How Teams Decide Human Oversight in AI-Powered Systems: A Mixed-Methods Study

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Lessons from Literature-Based Case Analysis

To contextualize the conflicting system objectives identified in the initial literature review, we analyzed a diverse set of real-world case studies drawn from both academic literature and formal media and industry sources. These examples highlight recurring conflicts between human oversight and competing system goals. Given the limited access to AI solutions in proprietary systems, literature-based and media-documented cases, which are not always peer-reviewed sources, offer valuable insights into the practical challenges encountered during the implementation of these AI solutions. Examining reported real-life cases enables us to examine how different domains implement, adapt, or struggle with human-oversight mechanisms. The insights derived from these cases inform and refine our proposed framework, offering concrete illustrations of oversight dilemmas and potential resolution strategies.

Autonomous Vehicles: Self-driving technologies developed by Tesla and Uber demonstrate the risk of full automation in safety-critical environments. Accidents involving autonomous vehicles have underscored the need for human-in-the-loop mechanisms, such as driver fallback systems or remote human intervention, to handle complex, unpredictable conditions (National Transportation, 2019, The Tesla Team, 2024, Gautam et al., 2024).

Fraud Detection in Banking: AI systems effectively detect fraudulent activity, but may trigger false positives, leading to legitimate transactions being blocked. In these cases, human analysts are needed to verify and override AI decisions, ensuring efficiency without compromising user experience (Teradata, Wallny et al., 2022, Fallen et al., 2025).

Medical Diagnostics: AI models have outperformed doctors in several specific tasks, such as cancer detection, but their black-box nature undermines trust. Clinicians require explainable outputs before acting on AI recommendations. This requirement was addressed by integrating explainable AI (XAI) tools, like saliency maps, but these often impose performance costs and still fall short of full transparency (Kolarik et al., 2023, Kelly et al., 2019).

AI in Hiring: Many companies including, for example, Amazon and LinkedIn, use AI for resume screening and behavior prediction. However, the AI models used for this purpose have limited transparency, raising fairness and accountability concerns. Organizations are therefore forced to choose between simplifying models for explainability or providing post-hoc rationalizations with limited fidelity (Dastin, 2018, Turvill, 2025, Zappulla, 2024).

Mortgage Approvals: Fairness-efficiency trade-offs are apparent in AI-assisted lending. Implementing fairness measures, such as demographic parity or equal opportunity, can help reduce systemic biases and promote equitable access to credit, especially for historically disadvantaged groups. However, such measures may lead to the approval of applicants who do not meet financial risk criteria, potentially increasing default rates and resulting in financial losses for lenders, or reject applicants who are financially qualified but fall outside the fairness criteria. This highlights the tension between fairness goals and financial optimization (Buijsman, 2024).

Predictive Policing: Policing systems, such as PredPol (Predictive Policing), identify high-risk areas using historical crime data. These tools can reinforce racial biases and lead to over-policing. Cities government authorities have responded with human oversight mechanisms and community input, but this measure slows AI responsiveness and adds operational complexity (Lau, 2020, Richardson et al., 2019, Deloitte, Clara, 2025).

AI for Pandemic Triage: During the COVID-19 pandemic, hospitals deployed AI models to support triage and resource allocation. Although these tools improved efficiency, most of them were trained on narrow datasets and risked biased outcomes. Experts emphasized the need for clinician oversight, contextual judgment, and ethical review to ensure fairness and accountability (Magnus et al.).

Aerospace Predictive Maintenance: AI enhances operational efficiency through predictive maintenance, reducing downtime and costs. However, human validation remains critical. Technicians reviewing AI-generated alerts reduce false positives and ensure safety in high-risk environments, reinforcing the value of structured oversight (Gautam, Patil, 2023).

Across sectors, common contradictions emerge: high-performance models struggle with explainability; fairness often demands manual intervention; and real-time systems face trade-offs between responsiveness and oversight. Human-in-the-loop mechanisms improve trust and accountability but introduce operational costs and design complexity, and at times potential human errors. These case studies reaffirm the need for domain-specific, ethically grounded oversight strategies that support both performance and legitimacy.

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